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Amenability and its Applications in the Banach-Tarski Paradox and Thompson's Group F

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Abstract

In this paper we will discuss amenability, its importance in the proof of the Banach-Tarski paradox, as well as its implications for Thompson's Group F. We will first introduce the concept of measure and amenability then give definitions and examples of each. Then we will write out the proof of the Banach-Tarski paradox and explain the role amenability plays in it. Lastly, we will introduce Thompson's Group F and discuss the open question of its amenability.

Contents

1 Introduction

The Banach-Tarski paradox is a fascinating subject in mathematics. Put simply, we can take a ball in 3-D (interior included), split it up into a bunch of different pieces, rotate and shift these pieces in a particular way, then reassemble these pieces to form 2 balls of the original size without adding anything! Admittedly, we can't do this in our physical world as everything is composed of a finite amount of molecules, whereas our ball is composed of an infinite amount of points. Turns out, when infinity comes into play, things become really weird[1].

In the study of Group Theory, amenability is a way to categorize groups into whether or not we can define a valid notion of "volume" on the group that doesn't change as we shift things around. One might begin to see the connection between amenability and the paradox!

In this paper we will discuss the concept of amenability, commonly seen in Group Theory to categorize groups. We will go through the definition of amenability (as well as the related definitions of measure) then give some examples. Then, we will investigate the importance of amenability and how it connects to logical paradoxes such as the Banach-Tarski paradox. Lastly, we will pose the open question of whether or not Thompson's Group F is amenable and outline several ongoing research methods.

The various definitions of measure and amenability vary in terms of detail and area of application. For our intents and purposes, we will talk about amenability on discrete groups using the probability measure.

This paper is expository in nature with the main goal of clearly explaining sources found elsewhere. My hope is that people (such as myself a few weeks ago) who have trouble understanding sources elsewhere can refer to this paper to have an intuitive understanding.

[1] Trust me.

2 Preliminary Definitions on Amenability

2.1 Measure and Amenability

In this paper, we will explore the notion of amenability, how it applies to groups, and why it is important. To do so, we must first understand measure, which is the foundation of amenability.

Intuitively, a measure can be seen as a notion of volume. A measure μ on a set X maps a subset A of X to a nonnegative number $\mu(A)$, which is called the measure of A. For the scope of this paper, I will use the probability measure, where the measure μ on a set X maps a subset A of X to a real number $\mu(A)$ in $[0, 1]$. Given a subset A of G, the probability measure can be thought of as answering the question: what is the probability that a random element of G is in A?

In general, a measure μ is a nonnegative, extended real-valued function defined on a collection of measurable subsets of X. We require that the measurable sets form a $\sigma{-}algebra$, meaning that complements, countable unions, and countable intersections of measurable sets are measurable. However, in the scope of this paper, we will be dealing with discrete groups, on which the probability measure defined on the power set (the set of all subsets of X) of the group suffices.

To read more about the general notion of a measure and more broader definitions, feel free to check out this chapter of Hunter's book.[10]

Here are some common and important measures defined below:

Definition 2.1 (Counting Measure)**.** Given a finite set X and a subset $A \in X$, the counting measure $\#$ on X is defined by $\#(A) =$ the number of elements in A.

Definition 2.2 (Probability Measure)**.** Given a set X and a subset $A \in X$, the probability measure on X is a finitely additive set function $\mu : A \to [0, 1]$ satisfying $\mu(\varnothing) = 0$ and $\mu(X) = 1$. i.e., if $A_n \in \mathcal{P}(X)$ and $\cup_{n\geq 1} A_n \in \mathcal{P}(X)$, then

$$\mu\left(\bigcup_{n\geq 1} A_n\right) = \sum_{n\geq 1} \mu(A_n)$$

for finite n.

Example 2.3. All finite groups admit a probability measure $\mu(A) : \mathcal{P}(G) \to [0, 1] := \frac{|A|}{|G|}$, where we take the measure of subset A of group G.

Definition 2.4 (Lebesgue Measure)**.** [7] If a set $X = \mathbb{R}$, a natural measure is the Lebesgue measure $\lambda(A)$, which returns the volume of a subset $A \in X$. i.e.,

$$\lambda(A) = \int_A 1 dx$$

Proposition 2.5. *[2][Properties of Measures] For any measure, it has several properties that are worth noting. Given a set X, its power set $\mathcal{P}(X)$, and an associated measure μ, we have*

1. *If $E, F \in \mathcal{P}(X)$ and $E \subseteq F$, then*

$$\mu(E) \leq \mu(F).$$

2. *If $E, F \in \mathcal{P}(X)$, then*

$$\mu(E \cup F) + \mu(E \cap F) = \mu(E) + \mu(F).$$

3. *If $\{E_n\}$ is any sequence in $\mathcal{P}(X)$, then*

$$\mu\left(\bigcup_{n \in \mathbb{N}} E_n\right) \leq \sum_{n \in \mathbb{N}} \mu(E_n).$$

4. *If $E_1 \subseteq E_2 \subseteq \cdots$ is a nested sequence in $\mathcal{P}(X)$, then*

$$\mu\left(\bigcup_{n \in \mathbb{N}} E_n\right) = \sup_{n \in \mathbb{N}} \mu(E_n).$$

5. *If $E_1 \supseteq E_2 \supseteq \cdots$ is a nested sequence in $\mathcal{P}(X)$ and at least one E_n has finite measure, then*

$$\mu\left(\bigcap_{n \in \mathbb{N}} E_n\right) = \inf_{n \in \mathbb{N}} \mu(E_n).$$

We can now formally define amenability as follows:

Definition 2.6. [4] A Group G is *amenable* if there is a left-invariant probability measure2.2 μ on G. That is, G is amenable if there is a probability measure μ such that $\mu(gA) = \mu(A)$ for all $g \in G$ and all subsets $A \in G$.

To familiarize ourselves with the notion of amenability, we introduce the following corollary:

Corollary 2.7. *1. All finite groups are amenable.*

Proof. Using the probability measure on a finite group G as defined in 2.3, simple verification shows that this probability measure is left-invariant. □

Finite groups are not the only ones that are always amenable. In fact:

Corollary 2.8 (*cont*). *1. All subgroups and quotients of amenable groups are amenable.*

2. Any extension of an amenable group by an amenable group is amenable.

3. Any direct union of amenable groups are amenable.

4. All abelian groups are amenable.

Proof. [8] We prove the first point: let G be an amenable group and μ its measure satisfying amenability. Using the Axiom of Choice, any subgroup H of G has a right traversal M in G (i.e. a group containing one "representative" of each coset of H in G). Thus, we can define a measure $\nu : \mathcal{P}(H) \to [0,1]$ by $\nu(A) := \mu(AM)$, which is easily verified to be a measure satisfying amenability. We can think of ν as expanding $A \in H$ "back" to an equivalent subset in G, then assigning the measure of that subset back to A.

For any quotient G/N of G, we use a similar logic and define $\lambda : \mathcal{P}(G) \to [0,1]$ by $\lambda(A) := \mu(AN)$, which is easily verified to be a measure satisfying amenability.

The proof for 2, 3, and 4 require broader notions of amenability and measure that are out of the scope of this paper. Interested readers are welcome to check out page 6 of Garrido's paper [8]. □

2.2 Paradoxical Decomposition

As alluded to in the introduction, amenability plays an important role in the proof of the Banach-Tarski paradox. However, when working with the groups used in the Banach-Tarski paradox, we would benefit from a different but equivalent definition of amenability, namely a paradoxical decomposition.

Definition 2.9. [19] Let G be a group acting on a set X and suppose $E \subseteq X$ is a nonempty subset of X. Then E is G-paradoxical *or paradoxical with respect to G* if there exist some positive integers r, s such that

$$\begin{aligned} E &= U_1 \sqcup \cdots \sqcup U_r \sqcup V_1 \sqcup \cdots \sqcup V_s \\ &= g_1U_1 \sqcup \cdots \sqcup g_rU_r = h_1V_1 \sqcup \cdots \sqcup h_sV_s. \end{aligned}$$

with $g_1, \cdots, g_r, h_1, \cdots, h_s \in G$.

Corollary 2.10. *In the above definition, if G acts on itself and $E = G$, we say that G has a paradoxical decomposition.*

Theorem 2.11. *Let G be a group. Then the following conditions are equivalent:*

1. *G is amenable, i.e. G has a left invariant probability measure.*

2. *G is not G-paradoxical, i.e. G does not have a paradoxical decomposition.*

3. *There does not exist a free action of G on a set X which is paradoxical.*

We will first prove that statements 1 and 2 are equivalent. This paper will cover the straightforward forward direction. The backwards direction is fairly complex and out of the scope of this paper. Readers who are interested in the backwards direction should check out these Cornell lecture notes where the approach of the proof is outlined [13].

Proof. Suppose that G has a paradoxical decomposition. Assume we can define a probability measure μ on G. Then we have

$$\mu(U_1) + ... + \mu(U_r) + \mu(V_1) + ... + \mu(V_s) = \mu(G) = 1.$$

Since μ is left-invariant, we have

$$\begin{aligned} 1 = \mu(G) &= \mu(g_1U_1) + ... + \mu(g_rU_r) + \mu(h_1V_1) + ... + \mu(h_sV_s) \\ &= \mu(G) + \mu(G) \\ &= 2\mu(G) = 2. \end{aligned}$$

which leads to a contradiction. Thus, it is impossible for us to define such a measure, meaning that G is non-amenable. $\square$

We will then prove that statements 2 and 3 are equivalent. Essentially, we need to prove that given a group G acting freely on a set X, G is paradoxical if and only if its free action on X is paradoxical; i.e., the free action admits a paradoxical decomposition of X. The idea of the proof is taken from these UTA lecture notes [11]. My proof corrects some important notation errors found in the notes.

Proof. We start with the forward direction. Let $G = A_1 \sqcup A_2 \sqcup \cdots \sqcup A_n \sqcup B_1 \sqcup B_2 \sqcup \cdots \sqcup B_m = \bigsqcup_{i=1}^{n} g_iA_i = \bigsqcup_{j=1}^{m} h_jB_j$ be a paradoxical decomposition of G. Assume G acts freely on a set X. We can partition X into a disjoint union of all its orbits with respect to G since G acts freely on X. Using the Axiom of Choice, we can select one member from each orbit and form a set M. We have that $\bigsqcup_{g\in G} gM$ is a disjoint partition of X. We now define

$$\hat{A}_i = \bigsqcup_{g\in A_i} gM, \qquad \hat{B}_j = \bigsqcup_{g\in B_j} gM.$$

We can see that these two sets are disjoint. On the other hand, since G has a paradoxical decomposition, we have

$$\begin{aligned} X &= \bigsqcup_{i=1}^{n} \hat{A}_i \sqcup \bigsqcup_{j=1}^{m} \hat{B}_j \\ &= \bigsqcup_{i=1}^{n} g_i \hat{A}_i = \bigsqcup_{j=1}^{m} h_j \hat{B}_j. \end{aligned}$$

Thus X is G-paradoxical.

We now move on to the backwards direction. Let $X = A_1 \sqcup A_2 \sqcup \cdots \sqcup A_n \sqcup B_1 \sqcup B_2 \sqcup \cdots \sqcup B_m = \bigsqcup_{i=1}^{n} g_i A_i = \bigsqcup_{j=1}^{m} h_j B_j$ be a paradoxical decomposition of X with respect to G. We can partition X into a disjoint union of all its orbits with respect to G since G acts freely on X. We then find that we can paradoxically decompose each orbit $\mathcal{O}$:

$$\begin{aligned} \mathcal{O} &= \bigsqcup_{i=1}^{n} (A_i \cap \mathcal{O}) \sqcup \bigsqcup_{j=1}^{m} (B_i \cap \mathcal{O}) \\ &= \bigsqcup_{i=1}^{n} (g_i A_i \cap \mathcal{O}) = \bigsqcup_{j=1}^{m} (g_j B_j \cap \mathcal{O}) \end{aligned}$$

Thus $\mathcal{O}$ is G-paradoxical. Since the group action of G on $\mathcal{O}$ is by nature transitive, we can use the following proposition:

Proposition 2.12. *If G acts transitively on a set C_a, then $C_a \cong G/G_a$ where $G_a = Stab_G(x)$ for some $x \in C_a$.*

The proof of this proposition is fairly straightforward and is found on page 114 of Dummit and Foote's Abstract Algebra textbook[6].

Since G acts freely on X, $G_a = \{e\}$ and $G \cong G/G_a$. Thus $\mathcal{O} \cong G$ and G has a paradoxical decomposition. □

The equivalence between these definitions is the foundation of the usage of amenability in the proof of the Banach-Tarski paradox. A simple example showing this equivalence is the free group of rank 2, F_2. This group can be seen as every possible word (string of letters) formed by concatenating the four characters a, b, a^{-1}, b^{-1} with each other and deleting any occurrence of $aa^{-1}, a^{-1}a, bb^{-1}$, and $b^{-1}b$. For a more thorough explanation of F_2, readers should feel free to check out page 12 of the Office Hours textbook [5]. Readers should also look at Figure 1 for an intuitive understanding of F_2.

Example 2.13. F_2 is non-amenable since it has a paradoxical decomposition2.11.

Proof. [12] Let F_2 be the free group on two generators a and b with identity element e. Let A^+ be the set of all elements in F_2 whose leftmost letter in reduced form is a, let A^- be the set of all elements in F_2 whose leftmost entry in reduced form is a^{-1}, and let B^+ and B^- be defined analogously. Let $C = \{e, a^{-1}, a^{-2}, ...\}$. Then we have that

$$\begin{aligned} F_2 &= (A^+ \cup C) \sqcup (A^- \setminus C) \sqcup B^+ \sqcup B^- \\ &= (A^+ \cup C) \sqcup a(A^- \setminus C) \\ &= B^+ \sqcup bB^- \end{aligned}$$

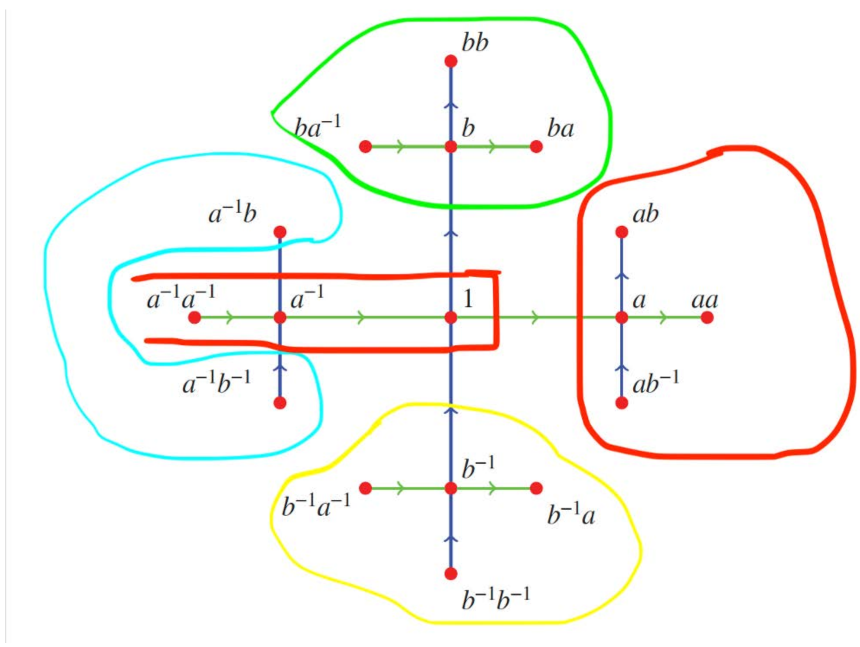


Figure 1: The Partition of F_2

where $(A^+ \cup C)$ is the red portion of the graph, $(A^- \setminus C)$ the blue portion, B^+ the green portion, and B^- the yellow portion. Intuitively, we can "shift" the green portion down by one unit by adding a b^{-1} in front of all the words. b becomes $b^{-1}b = 1$, ba becomes $b^{-1}ba = a$, and everything will have been translated down one unit. These words after translation are in fact all the words in F_2 except those that start with b^{-1}, because that would imply an occurrence of bb^{-1} in the original word which is impossible. Combine this with the yellow portion, and we have the whole graph! We can do the same process by shifting the blue portion right one unit and combining it with the red portion.

Suppose then that we did in fact have a measure μ satisfying all the properties of amenability. Then we would have

$$\begin{aligned}\mu(F_2) &= \mu(A^+ \cup C) + \mu(A^- \setminus C) + \mu(B^+) + \mu(B^-) \\ &= \mu(A^+ \cup C) + \mu(a(A^- \setminus C)) + \mu(B^+) + \mu(bB^-) \\ &= 2\mu(F_2).\end{aligned}$$

□

When originally reading this Vanderbilt source, I was confused as to why the author partitioned the group F_2 in this lopsided way, defining the set C and removing it from one partition and adding it to another. I didn't understand why the author didn't simply just split F_2 into e, A^+, A^-, B^+, and B^-. However, upon reflection, if we tried to do the proof in the same method with this partition, we would end up with an inequality, namely

$$\begin{aligned}\mu(F_2) &= \mu(e) + \mu(A^+) + \mu(A^-) + \mu(B^+) + \mu(B^-) \\ &= \mu(e) + \mu(A^+) + \mu(aA^-) + \mu(B^+) + \mu(bB^-) \\ &\leq 2\mu(F_2).\end{aligned}$$

since we never accounted for the origin (1) in our subdivisions. This still supports the proof! However, the idea of the Banach-Tarski paradox, put simply, is the notion of creating two identical copies of an object from the object itself. Having strict equality here is a good supporting example as to how it works in 2D. We will later see how this technique is mirrored in the proof of the Banach-Tarski paradox.

3 The Banach-Tarski Paradox and the Importance of Amenability

Amenability is important in many areas of Mathematics, an important one being the Banach-Tarski paradox. Amenability lies in the core of the proof of the Banach-Tarski paradox.

The Banach-Tarski paradox has many forms–for our intents and purposes, we start with the following definition:

Definition 3.1. A ball in $\mathbb{R}^3$ is equidecomposable with two copies of itself.

We will later show that the Banach-Tarski paradox holds for any two bounded subsets with nonempty interior, not just balls!

Definition 3.2. Two sets X and Y are said to be equidecomposable if you can partition set X into a finite number of subsets and reassemble them (by rigid motions, i.e. rotations and translations only) to form set Y.

To strengthen our understanding of equidecomposability, consider the following example:

Proposition 3.3. *A circle is equidecomposable with a circle missing a single point.*

Proof. We will prove the case for the unit circle. It is clear that the same process can be done for any circle after adjustment for size. Suppose we have a unit circle S^1 missing the point $(1, 0)$. We can choose any missing point here, since $S^1 \setminus x$ for any x other than $(1, 0)$ is clearly isometric to $S^1 \setminus (1, 0)$. Let $A = \{(\cos n, \sin n) | n \in \mathbb{N}\}$. Since π is irrational and n is rational, we don't have any duplicate points in A. Let $B = S^1 \setminus (1, 0) \setminus A$.

We rotate A about the origin by -1 radians. Call the resulting set A'. The rotation maps $(\cos 1, \sin 1)$ to $(1, 0)$, $(\cos 2, \sin 2)$ to $(\cos 1, \sin 1)$, and so forth. For any $(\cos m, \sin m)$, it is mapped to by $(\cos m + 1, \sin m + 1)$. Thus every point originally in A is still in A', i.e. $A' = A \sqcup (1, 0)$. Thus $S^1 \setminus (1, 0) = A \sqcup B$ is equidecomposable with $S^1 = A' \sqcup B$, i.e. the unit circle missing a single point is equidecomposable with the unit circle. $\square$

We can think of the Banach-Tarski paradox as such: we can take a ball, divide it up, and recombine the pieces in a way such that we can form 2 balls out of those parts. Remember how the free group F_2 could be divided and regrouped to form 2 copies of itself? We will follow that approach: we will try (and succeed) to find a group isomorphic to F_2 for our ball, have this group act on our ball, subdivide the infinite amount of points composing the ball into portions similar to that of F_2 we did earlier, and duplicate all these points in a similar way we did F_2.

3.1 Proof of the Banach-Tarski Paradox

I will largely draw from Robinson's proof[14].

Our first step is to then define this group. We define the free group of rotations about the origin in $\mathbb{R}^3$ with two generators. Note that we will explore the case of the unit ball centered at the origin (interior included), which we will call $L : \{v \in \mathbb{R}^3 | \|v\| \leq 1\}$. Without loss of generality this method can be used with a ball of any size centered at any point. We will show this later.

Choose $\theta = \arccos \frac{1}{3}$ and let A be a rotation by θ about the x axis, and let B be a rotation by θ about the z axis. We can also express A and B in matrix form using the rotation matrix:

$$A = \frac{1}{3}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & -2\sqrt{2} \\ 0 & 2\sqrt{2} & 1 \end{pmatrix}, \qquad A^{-1} = \frac{1}{3}\begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 2\sqrt{2} \\ 0 & -2\sqrt{2} & 1 \end{pmatrix},$$

$$B = \frac{1}{3}\begin{pmatrix} 1 & -2\sqrt{2} & 0 \\ 2\sqrt{2} & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \qquad B^{-1} = \frac{1}{3}\begin{pmatrix} 1 & 2\sqrt{2} & 0 \\ -2\sqrt{2} & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

since $\sin(\arccos \frac{1}{3}) = \sqrt{1 - (\frac{1}{3})^2} = \frac{2\sqrt{2}}{3}$ and $\cos(\arccos \frac{1}{3}) = \frac{1}{3}$.

Let $\mathcal{G} =< A, B >$ be the group generated by A and B. Note that we choose $\theta = \arccos \frac{1}{3}$ since it is an irrational multiple of π. We could choose any irrational multiple of π; $\arccos \frac{1}{3}$ is just an example. We have no equivalence relations in $\mathcal{G}$, meaning that no two reduced sequences of rotations are equal to each other.

Proposition 3.4. *$\mathcal{G}$ is a free group, i.e. $\mathcal{G} \cong F_2$.*

Proof. [14] We prove by contradiction. Suppose there exists some $g \in \mathcal{G}$ such that $g(0,1,0) = (0,1,0)$. We choose $(0,1,0)$ as the unit vector in the y direction is a simple example to work with.

First note that $g(0,1,0)$ is necessarily of the form $\frac{1}{3^n}(a\sqrt{2}, b, c\sqrt{2})$, with $a, b, c \in \mathbb{Z}$ and n being the reduced word length of g. We show this by induction. Note that $(0,1,0)$ is clearly in this form. Then we suppose $g'(0,1,0) = \frac{1}{3^{n-1}}(a\sqrt{2}, b, c\sqrt{2})$ where $g = Ag'$ or $A^{-1}g'$ or Bg'

or $B^{-1}g'$. $g(0,1,0)$ is then necessarily in one of the following forms:

$$\begin{aligned}
Ag'(0,1,0) &= \frac{1}{3^n}(3a\sqrt{2}, b-4c, (2b+c)\sqrt{2}); \\
A^{-1}g'(0,1,0) &= \frac{1}{3^n}(3a\sqrt{2}, b+4c, (-2b+c)\sqrt{2}); \\
Bg'(0,1,0) &= \frac{1}{3^n}((a-2b)\sqrt{2}, 4a+b, 3c\sqrt{2}); \\
B^{-1}g'(0,1,0) &= \frac{1}{3^n}((a+2b)\sqrt{2}, -4a+b, 3c\sqrt{2}).
\end{aligned}$$

which are all of the desired form.

In order for $g(0,1,0) = (0,1,0)$ with $g(0,1,0)$ of the form $\frac{1}{3^n}(a\sqrt{2}, b, c\sqrt{2})$, we must have $a = c = 0$ and $b = 3^n$ for $n > 0$. Therefore $a \equiv b \equiv c \equiv 0 \mod 3$. It can be shown by induction and listing all possible results of applying A and B modulo 3 that this is not possible. The details are lengthy but not very interesting or elegant, so they will be omitted here. $\square$

With our group defined, we can now decompose the ball in a similar fashion as we did with the free group F_2 previously. We will group (almost) every point (apart from the center) into a particular orbit of $\mathcal{G}$.

Definition 3.5. Two points, a and b, belong to the same *orbit* if and only if there exists some g in G such that $g(a) = b$.

Note that intuitively, this is exactly "mapping" a group of F_2 on the ball. One orbit in G can be seen as one copy of F_2 covering an infinite amount of points on the ball. Intuitively, we want to first find all the orbits of $\mathcal{G}$ on the unit ball without its center, which we will call $L' = L \setminus e$ (since the center doesn't shift when rotated). Then, using the Axiom of Choice, we choose a "starting point" from each orbit and make them form a set M. Then, almost all points in the ball can be reached from M by group actions in $\mathcal{G}$ (i.e. rotations). We can think of the ball as being composed of $|M|$ copies of F_2, with each orbit being a copy, and analogously, we can do the decomposition and recomposition of F_2 for every orbit and duplicate (most) of the ball like this. Notice that all the orbits in $\mathcal{G}$ are disjoint, i.e. every point on the ball is contained in exactly one orbit of $\mathcal{G}$ (or copy of F_2).

Proposition 3.6. *Let $\mathcal{O}_1(x_1)$ and $\mathcal{O}_2(x_2)$ be two distinct orbits of $\mathcal{G}$. Then $\mathcal{O}_1 \cap \mathcal{O}_2 = \varnothing$.*

Proof. We prove the contrapositive. Suppose there exists an element $o \in \mathcal{O}_1(x_1) \cap \mathcal{O}_2(x_2)$. This means $o = g_1 x_1 = g_2 x_2$ for $g_1, g_2 \in \mathcal{G}$. Then $x_1 = g_1^{-1} g_2 x_2$. Then $x_1 \in \mathcal{O}_{\in}(x_2)$. Thus $\mathcal{O}_1(x_1) \subset \mathcal{O}_2(x_2)$. Using the same logic but in reverse $\mathcal{O}_2(x_2) \subset \mathcal{O}_1(x_1)$. Thus $\mathcal{O}_1(x_1) = \mathcal{O}_2(x_2)$. $\square$

When doing this procedure, we want to make sure that each point is reached by a unique sequence of rotations from a particular starting point in M (so that $\mathcal{G}$ is isomorphic with F_2). This means we need to rule out a special set of points, namely fixed points of rotation (which we will call D). Every element of $\mathcal{G}$ generated by A and B can be viewed as a rotation by an axis through the center of the ball. For example, the rotation AB are two rotations completed successfully, but we can also view it as a singular rotation. Points on an axis of a rotation do not move when the rotation is completed; we have a countable number of elements in G, and we have a countable amount of points on a countable amount of rotation axes. These are our fixed points of rotation, and we can reach them in more than one way since there is an element other than the identity rotation (i.e. the element of $\mathcal{G}$) that fixes them. In our decomposition of the ball, we do not consider these points as otherwise our orbits cannot be seen as copies of F_2. Thus, we are really decomposing a ball without the center and it's fixed points of rotation, i.e. $L' \setminus D$.

Now we go about decomposing $L' \setminus D$. Analogously, we define C to be all points in $L' \setminus D$ that are reached by a sequence consisting of only A^- from M, i.e.

$$C = \bigcup_{i=1}^{\infty} A^{-i} M$$

Let $S(A)$ denote the set of all expressions in $\mathcal{G}$ that begin with A, after having been fully simplified using the group axioms, which in our case, is simply $AA^{-1} = A^{-1}A = BB^{-1} = B^{-1}B = e$. For example, AB is in $S(A)$, but $AA^{-1}B$ is not, because $AA^{-1}B = B$. We can think of $S(A)M$ as being M copies of the "right" portion in the graph of the free group F_2. Now define $S(A^{-1}), S(B)$, and $S(B^{-1})$ similarly. We then partition $L' \setminus D$ as follows: $L' \setminus D = P_1 \sqcup P_2 \sqcup P_3 \sqcup P_4$, where

$$\begin{aligned}
P_1 &= S(A)M \cup M \cup C \\
P_2 &= S(A^{-1})M \setminus C \\
P_3 &= S(B)M \\
P_4 &= S(B^{-1})M
\end{aligned}$$

Notice that this decomposition is exactly parallel to our decomposition of F_2, except this time we have $|M|$ copies of F_2. Analogously, we rotate P_2 by A, add it with P_1, and get

$L' \setminus D$. We can do the same with BP_4 and P_3. Thus we have:

$$L' \setminus D = P_1 \sqcup AP_2$$
$$L' \setminus D = P_3 \sqcup BP_4$$

We have effectively duplicated (most) of the ball analogously to F_2. Even through the equations, we can intuitively see that we're doing the same operations we did on F_2, just "multiplied" by M. We now move on to the rest of the points, namely the fixed points of rotation and the center point. We first show that:

Proposition 3.7. *$L' \setminus D$ and L' are equidecomposable.*

Proof. Since the points on D are on a countable number of axes, we can find an axis I through the center point that overlaps with none of the axes of D. Again, since there are a countable number of axes in D, we can find an angle θ such that no amount of rotations with I as axis of angle θ map some point on D to another point on D. We can visualize it like this: for any pair of axes in D, there might be an angle ϕ such that every point on one axis maps to every point on the other after a rotation of ϕ by I. There are at most a countable number of such ϕ. Thus, we can always find an angle θ such that no multiple of it equals the ϕ's we found.

We now define a set E as follows, with ρ being the rotation with I as axis of angle θ:

$$E = D \sqcup \rho(D) \sqcup \rho^2(D) \sqcup \rho^3(D) \sqcup \cdots$$

Clearly, $L' = E \cup (L' \setminus E)$, which is equidecomposable with $\rho(E) \cup (L' \setminus E)$. However, we can see that $\rho(E) = E \setminus D$. Thus we have $\rho(E) \cup (L' \setminus E) = (E \setminus D) \cup (L' \setminus E) = L' \setminus D$. Thus, we have that L' is equidecomposable with $L' \setminus D$. $\square$

We now move on to the center point. Recall that at the start of the chapter, we proved that a circle was equidecomposable with a circle missing a single point!3.3 Thus, we can simply take a circle S' within the unit ball that intersects the center point. Using 3.3, we have $L = S' \sqcup (L' \setminus S')$ is equidecomposable with $L' = S' \setminus (0,0) \sqcup (L' \setminus S')$. Thus, we have that the unit ball is equidecomposable with the unit ball missing its center point.

3.2 Recap of Proof

This proof is a fascinating one composed of taut logical links. I will go over the logic of the proof to strengthen our understanding:

1. Generalize bounded subsets with nonempty interior to balls in 3-D Euclidean Space
2. Generalize balls in 3-D Euclidean Space to the unit ball centered at the origin
3. Define rotation group $\mathcal{G}$ (isomorphic to F_2) that acts on the unit ball
4. Separate unit ball into 3 parts: center point, fixed points of the axes of rotation, and all other points (which we have called $L' \setminus D$)
5. Partition all points in $L' \setminus D$ into orbits of $\mathcal{G}$ (copies of F_2)
6. Use Axiom of Choice to select one point from each orbit to form a set M of starting points
7. Use a decomposition analogous to F_2 to duplicate all points in $L' \setminus D$ by treating $L' \setminus D$ as $|M|$ copies of F_2
8. Show that $L' \setminus D$ is equidecomposable with the unit ball

3.3 Connection to Amenability

To show the importance of amenability in the Banach-Tarski paradox, we turn to this other definition of the paradox:

Definition 3.8. [19] Any ball in $\mathbb{R}^3$ is paradoxical with respect to G_3, the group of isometries of $\mathbb{R}^3$.

If we look back to our equivalence definitions, namely statements 1 and 3 in 2.11, we find that showing G_3 is non-amenable automatically gives us a set X that is paradoxical with respect to a free action of G_3. We can see that our rotation group $\mathcal{G}$ defined in the paper is a non-amenable group as it is isomorphic to F_2. Since $\mathcal{G}$ is a subgroup of G_3, G_3 is non-amenable, thus we have a set X that is paradoxical with respect to a free action of G_3. The entire proof is really showing that this set X can be the unit ball; what we have done is showing that the unit ball is paradoxical with respect to the free action $\mathcal{G}$. This is useful since the case of the Banach-Tarski paradox with the unit ball can be extended to any bounded subsets in $\mathbb{R}^3$, which gives us the stronger version of the paradox:

Theorem 3.9. *Any two bounded subsets of $\mathbb{R}^3$ with nonempty interior are G_3-equidecomposable, i.e. the two sets are equidecomposable with respect to the free actions of G_3*

Proof. [16] Let E be a bounded subset of $\mathbb{R}^3$ with nonempty interior. To prove the theorem, we simply need to show that E is G_3-equidecomposable with the unit ball. It is clear that we can find a ball B in $\mathbb{R}^3$ which is contained in E and another ball B' in $\mathbb{R}^3$ which contains E. It is clear that B is G_3-equidecomposable with a subset of E and that E is G_3-equidecomposable with a subset of B'. If we show that all balls in $\mathbb{R}^3$ are G_3-equidecomposable with each other, then it follows that E is G_3-equidecomposable with the unit ball. This leaves us to prove the proposition:

Proposition 3.10. *Any two balls in* $\mathbb{R}^3$ *are* G^3*-equidecomposable.*

The proof on page 13 of Shapiro's paper is quite straightforward [16]. For brief exposition, we need only to prove that the unit ball is equidecomposable with another ball. Without loss of generality, assume that the other ball has larger radius (if smaller, simply reverse the argument). We can "cover" this ball with finitely many unit balls as it has finite volume. We then "disjointify" these unit balls by getting rid of the areas of overlap. Each partition is then equidecomposable with the unit ball. The Banach-Tarski paradox shows that a unit ball is equidecomposable with two copies of itself, thus we can do the "Banach-Tarski iteration" finitely many times to show that a unit ball is equidecomposable with finitely many copies of itself. Thus we have the bigger ball being equidecomposable with a subset of the union of the cover of finitely many unit balls, which is equidecomposable with the union of the finitely many disjointed partitions, which is equidecomposable with the finitely many unit balls, which is finally equidecomposable with the unit ball. □

There is another interesting result regarding the Banach-Tarski paradox, which is:

Proposition 3.11. *The Banach-Tarski paradox holds for all dimensions* $n \geq 3$*, but not for* $n = 2$*.*

Proof. [19] Readers should visit page 78 of Wagon's textbook and read the proof presented there in conjunction with this proof.

We start by proving the first statement. We have already proved the $n = 3$ case, and we can prove all cases $n > 3$ using induction. Note that for the $n = 3$ case we have "grouped" all the points forming the ball according to their orbits with respect to $\mathcal{G}$. For $n = 4$, we keep the same groupings, and we group the points forming the hypersphere (4d ball) using our original grouping excluding the poles ($(0, 0, 0, \pm 1)$). More specifically, we group (x_1, x_2, x_3, z) based on which group (x_1, x_2, x_3) belongs to. With all our points for our ball in $\mathbb{R}^4$ grouped, we define the new rotations using our two original rotations. More specifically, we use the exact the same rotations, except we also want them to fix z. i.e., we have

$$A' = \frac{1}{3}\begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & -2\sqrt{2} & 0 \\ 0 & 2\sqrt{2} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad A'^{-1} = \frac{1}{3}\begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 2\sqrt{2} & 0 \\ 0 & -2\sqrt{2} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

with B' and B'^{-1} defined analogously.

Note that with this grouping of points forming the hypersphere and these new rotations, we can paradoxically decompose the hypersphere without its poles in the exact same way we did the ball in $\mathbb{R}^3$! Using 3.3, we can "ignore" the 2 poles. Simply repeating this procedure yields the paradoxical decomposition in higher dimensions. In essence, the higher-dimensional ball (minus poles) is treated like a continuous stack of lower-dimensional balls, each inheriting the same paradoxical decomposition.

For the second statement, we quote from Wagon and use 2.11: "The isometry group of the plane does not contain a free noncommutative subgroup, and so the construction of the Banach–Tarski Paradox does not work in the plane, or in the line." □

Lastly, some readers might wonder the importance of the probability measure in our discussion. What if we used some other measure such as the Lebesgue measure? Recall that, using 2.11, 3.8 is equivalent to saying that it is impossible to assign a finitely additive, left-invariant, nonzero measure to every subset of the ball. We can see that our definition of amenability is measure-dependent. Throughout this paper, our definition of amenability has been reliant on the properties of the probability measure, i.e. it being finitely additive, left-invariant, and nonzero (when measuring nonempty subsets). If we were to change the measure, our definition of amenability would have to change with it, so would our definition of the paradox. The Banach-Tarski paradox can also be seen as an impossiblity of the concurrent satisfaction of the following conditions:

1. We can define a measure on every subset of the unit ball in $\mathbb{R}^3$.

2. This measure is finitely additive.

3. This measure is left-invariant.

4. This measure is nonzero for every nonempty subset of the ball.

3.4 Axiom of Choice

In numerous locations of the proof, such as step 6 of the recap of the proof, we used an important tool that we have glossed over this entire paper: the Axiom of Choice. There are numerous definitions of it; for our intents and purposes, this definition from a Vanderbilt site suffices:

Definition 3.12. [15] Let C be a collection of nonempty sets. Then we can choose a member from each set in that collection. In other words, there exists a choice function f defined on C with the property that, for each set S in the collection, $f(S)$ is a member of S.

This seems intuitive; if we have nonempty sets, then obviously we can select a member from each set. However, defining *how* to select these members turns out to be extremely difficult (and impossible) in some cases. For finite sets, this can be straightforward, with an example being simply selecting the first member (or n^{th} member, provided the set has at least n members) of each set. However, when we have sets of infinite size, we might not be able to do the same. Consider the following example:

Example 3.13. [15] Let C be all nonempty subsets of the real line. So far, no one has ever found a suitable choice function f for this collection C.

Readers with some knowledge of Real Analysis might be able to intuitively see where things go wrong here. Subsets can get smaller and smaller, and it is unclear how we can select members mathematically from these subsets. If we were to look at the Banach-Tarski paradox, we are precisely dealing with uncountable sets like these. There are uncountably many points in $L' \setminus D$, and there are an uncountably many orbits of $\mathcal{G}$ on $L' \setminus D$, each with an uncountably many points. Simply stating the Axiom of Choice is not rigorous for some, as in this case, we cannot clearly define a choice function f that chooses the members. In fact, to quote from the Vanderbilt webpage, "most "ordinary" mathematicians – i.e., most mathematicians who are not logicians or set theorists – accept the Axiom of Choice chiefly because their work is simpler with the Axiom of Choice than without it." [15]

Readers interested in more about how the Axiom of Choice connects to the Banach-Tarski paradox are welcome to check out Wagon and Tomkowicz's book.[19] This book extensively discusses the Banach-Tarski paradox. Being a college senior, I often found that the content was very condensed and I had to do lots of additional readings in order to understand the book. Nevertheless, I do recommend it as a good source along with more expository sources.

Readers interested in an intuitive video explanation of the Banach-Tarski paradox are welcome to check out Vsauce's video on this topic. [18] I hope this video inspires you like it did me some many odd years ago.

4 Open Question: Is Thompson's Group F Amenable?

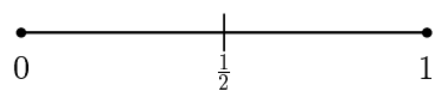
Amenability is important in many areas of maths. Thinking back to 2.11, with amenability, we have the equivalence definitions as well as a series of corollaries (for instance the non-inclusion of F_2 as a subgroup, having non-exponential growth, etc). Most groups fall neatly into one category or the other, like our rotation group above being non-amenable and having F_2 as a subgroup.

In the study of Group Theory, there is a very interesting group called the Thompson's Group F which encompasses many interesting paradoxical behaviours connected to other areas of Mathematics. Thompson's Group F has many traits often associated with amenability, and other traits often associated with non-amenability. The question of its amenability is still an open question to this day. Various attempts have been made to discern its amenability, and I will briefly show two angles in this section. I will largely draw from Belk's PhD Dissertation on this group for my definitions and expositions[1].

4.1 Definitions

Definition 4.1. A *homeomorphism* is a continuous function that is bijective and whose inverse is also continuous.

Thompson's group F is a certain group of piecewise-linear homeomorphisms of $[0, 1]$. To understand this intuitively, we take the unit interval $[0, 1]$. We can cut it in half to get the following:



We can "cut" certain subdivisions of the unit interval repeatedly in half to get a subdivision of the unit interval.

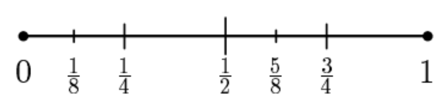


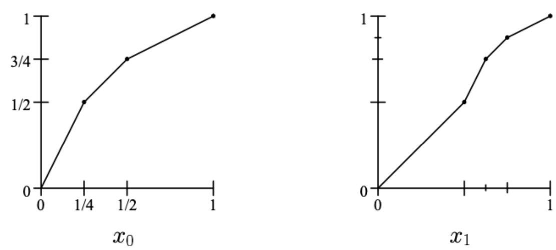
Figure 2: A dyadic subdivision

Any subdivision of the unit interval obtained in such a manner of repeatedly cutting intervals in half is called a *dyadic subdivision*.

Each interval of a dyadic subdivision is of the form $[\frac{k}{2^n}, \frac{k+1}{2^n}]$, where $k, n \in \mathbb{N}$. We call these the *standard dyadic intervals.*

We can now specify the definition of F. Given a pair D_1, D_2 of dyadic subdivisions with the same number of cuts (i.e., same number of intervals), we can define a piecewise-linear homeomorphism $f : [0,1] \to [0,1]$ by sending each interval of D_1 linearly onto the corresponding interval of D_2. We call this a dyadic rearrangement of $[0,1]$.

Example 4.2. Here are two examples of dyadic rearrangements which we will call x_0 and x_1:



We can see that for any dyadic rearrangement, it satisfies the following two properties:

1. All the slopes of f are powers of 2
2. All the breakpoints of f (point where f changes slope) have dyadic rational coordinates

Readers who are interested in the proof of these two properties should check out Belk's proof on page 12 of his dissertation [1].

Definition 4.3 (Thompson's Group F)**.** Thompson's Group F is the set F of all dyadic rearrangements with finitely many break points under function composition.

4.2 Group Attributes

Here are some facts about F that will aid our understanding of the group:

Proposition 4.4. *F is torsion free, i.e. every non-trivial element has infinite order.*

Proof. [5] Given any nontrivial element of F, consider the first point on the curve of the element with (right-sided) slope not equal to 1 (for instance, this point is $(0,0)$ for x_0 and $(\frac{1}{2}, \frac{1}{2})$ for x_1). Any nontrivial power of this element will have (right-sided) slope not equal to 1 at this point at the same x-value, meaning that it cannot be the identity. $\square$

Proposition 4.5. *F contains a subgroup isomorphic to $\mathbb{Z}^k$ for each $k \geq 0$.*

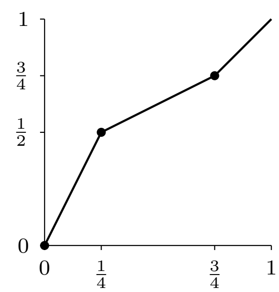
Before diving in this proof, we first introduce a definition:

Definition 4.6. We define the *support* of an element in F to be the complement of the fixed set, i.e. the "portion" of the curve that does not "match" the identity map. For instance, the support of x_1 is the part of its curve with x-values greater than $\frac{1}{2}$.

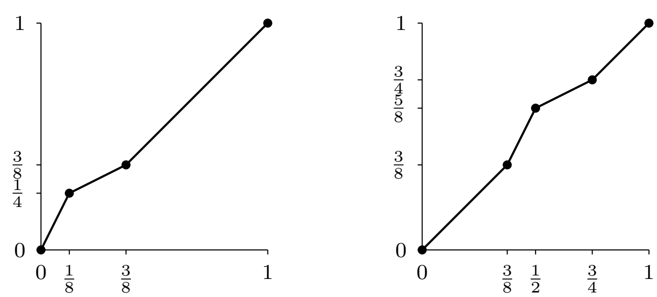
Proof. [5] It is not hard to see that elements of F with disjoint supports will necessarily commute (since every point only gets "shifted" once). Therefore, given any k, it is enough to find k elements with disjoint supports. They commute with each other while all having infinite order.

I will define a set of functions called "bump functions": elements in F that coincide with the identity for most of the interval, but deviate from the identity in a short interval. For these bump functions, the support is the short interval. In that short interval, the curve should have one line segment with slope greater than 1 and another line segment with slope less than 1. We define the set of k bump functions as follows:

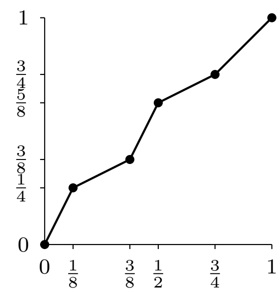
For $k = 1$, we only need one bump function. We define this bump function as



For $k = 2$, we simply "split" this bump function into two parts like so:



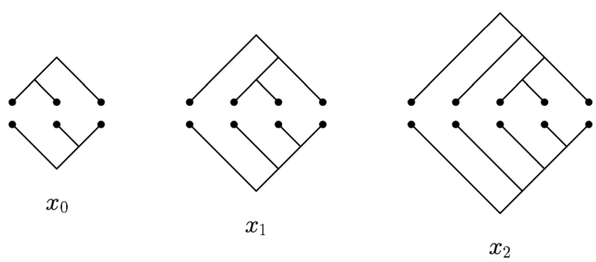
We can see that function composition of these two bump functions yield:



For $k = 3$, we simply split each of the bump functions in the $k = 2$ case in half and only "use" three of them. We can see that for any k, we can split the bump function in the $k = 1$ case into $2^n \geq k$ (for a natural number n) pieces and "use" k of those pieces to form our subgroup isomorphic to $\mathbb{Z}^k$. Simple verification yields that each of these bump functions are elements of F. $\square$

This analytical definition is one of many ways to represent F. There is namely another intuitive way to represent F using pairs of binary trees. This representation will significantly aid understanding of F; readers are highly recommended to check out page 4 of Belk's thesis for an overview[1].

Example 4.7. x_0, x_1, and x_2 in binary tree form:



To provide more context for the discussion of the amenability of F, I will show that F is finitely generated. We first introduce an infinite presentation of F:

Definition 4.8 (Infinite Presentation)**.** $F = \{x_0, x_1, \cdots | x_i^{-1} x_j x_i = x_{j+1}, i < j\}$

We will show that in fact:

Proposition 4.9. *The finite presentation* $F = \{x_0, x_1, | x_1^{-1} x_2 x_1 = x_3, x_1^{-1} x_3 x_1 = x_4\}$ *is equivalent to the infinite presentation above 4.8.*

Proof. Observe that the two relations in the finite presentation are examples of the relation described in the infinite presentation. We simply need to show that all x_i can be generated by x_0 and x_1.

We take the two relations as example. If we left multiply the first relation by x_0 and right multiply the first relation by x_0^{-1}, we get $x_0 x_1^{-1} x_2 x_1 x_0^{-1} = x_0 x_3 x_0^{-1}$. Observe that the relation described in 4.8 can be rewritten as $x_j x_i = x_i x_{j+1}$. Thus the first relation can be written as $x_0 x_1 x_0^{-1} = x_2$. Thus x_2, and therefore x_3 and all x_i, can be generated by x_0 and x_1. The reason we have two relations here is to "start" the recursive process of defining all x_i by x_0 and x_1. □

Thus, now that we know the two presentations are equivalent, it is left for us to show that the infinite presentation in fact generates F. Clay's textbook does a good job outlining the proof in detail, and I will refer readers to page 343 of his textbook [5] to read the proof as it is out of the scope of this paper.

4.3 Is F Amenable?

Let AG denote the class of all amenable groups, and let EG denote the smallest class of groups containing all finite and abelian groups and closed under subgroups, quotients, extensions, and direct unions (the elementary amenable groups). Let NF denote the class of all groups that do not contain a free subgroup of rank two [1].
According to 2.7 and 2.8, we have

$$EG \quad \subset \quad AG \quad \subset \quad NF$$

The question then arises as to whether these inclusions are proper. The amenability of F is investigated because F might lie in between these classes. We will show that $F \in NF/EG$, i.e. F does not contain a free subgroup of rank two, and it is not elementary amenable. The proof of the latter is out of scope of this paper. Readers are welcome to check out page 28 of Belk's dissertation [1].

Recall that the proof of the Banach-Tarski paradox relies on the isometry group G_3 being non-amenable as it contains F_2 as a subgroup. We showed above that F contains a $\mathbb{Z}^k$ for each $k \geq 0$. Would it contain F_2? We can see that this approach fails with F, as:

Proposition 4.10. *F does not contain the free group F_2.*

Proof. [1] Let $f, g \in F$. We need to show that f and g do not generate a free subgroup.

Assume first that f and g have no common fixed points in $(0,1)$. Observe then that any $x \in (0,1)$ can be mapped arbitrarily close to 0 using elements of $< f, g >$, as if this was not the case, the supremum of the orbit of x in $< f, g >$ would be common fixed point(s) of f and g. In particular, for any element $j \in < f, g >$, there exists an element

h such that we can push the support of j by conjugating it with h (i.e., $h^{-1}jh$) and have the support of $h^{-1}jh$ be completely disjoint from the support of j. More specifically, the support of j is defined as $x \in \text{supp}(j) \Leftrightarrow j(x) \neq x$. The support of $h^{-1}jh$ is defined as $x \in \text{supp}(h^{-1}jh) \Leftrightarrow h^{-1}jh(x) \neq x$, which means $jh(x) \neq h(x)$. Let $y = h(x)$. Thus $y \in \text{supp}(j) \Leftrightarrow x = h^{-1}(y) \in h^{-1}\text{supp}(j)$.

Thus we have:

$$\text{supp}(h^{-1}jh) = h^{-1}\text{supp}(j)$$

signifying us shifting the support of j using h^{-1}. Since there exists an element in $< f, g >$ to push any $x \in (0, 1)$ arbitrarily close to 0, there exists such a h^{-1} to shift the support of j arbitrarily close to 0 and away from the original support of j for any j. Thus $h^{-1}jh(x)$ commutes with j, meaning that f and g do not generate a free subgroup.

Now assume that f and g have common fixed points in $(0, 1)$. Then the support of $< f, g >$ is the union of the interiors of finitely many dyadic intervals $I_1, I_2, \cdots, I_n$, with the fixed points separating each interval. We can thus define an monomorphism (i.e. injective homomorphism) as follows:

$$< f, g > \hookrightarrow PL_2(I_1) \times PL_2(I_2) \times \cdots \times PL_2(I_n)$$

The image of $< f, g >$ in each $PL_2(I_i)$ is not free, since f and g don't have common fixed points in each dyadic interval I_i. Observe that we can define a surjective homomorphism between F_2 and $< f, g >$. Thus each composition

$$F_2 \twoheadrightarrow < f, g > \rightarrow PL_2(I_i)$$

has a nontrivial kernel. Thus the kernel of $F_2 \twoheadrightarrow < f, g >$ is the intersection of finitely many nontrivial subgroups of F_2 and is therefore nontrivial. Thus $F_2 \twoheadrightarrow < f, g >$ is not injective, meaning that f and g do not generate a free subgroup. $\square$

Even though F does not contain the free group F_2, it still has the following property:

Proposition 4.11. *F has exponential growth*[2].

This means we cannot directly discern its amenability just from its growth rate using the following proposition:

[2]The proof is out of the scope of this paper. Readers are welcome to check out page 49 of Burillo's book[3].

Proposition 4.12. *Any group with subexponential growth is amenable*[3].

4.3.1 Current Research on the Amenability of F

I will mostly be outlining results from Guba's paper[9].

As we can see, F possesses contradictory behaviours. It is not elementary amenable and has exponential growth, yet it does not contain F_2. Amenability has many equivalence definitions; researchers are working on some of them by analyzing the Cayley graph of F. We have already seen an example of a Cayley graph in 1, and more generally for the Cayley graph of any finitely generated group, vertices are group elements, and edges correspond to multiplication by generators. The Cayley graph of F does not have a straightforward visual representation like F_2; instead, researchers rely on other methods to understand the geometry of the graph, such as using forest-diagrams[1] and numerical methods.

Cheeger Isoperimetric Constant : For any finite set of vertices Y in a Cayley graph, we examine its Cheeger boundary $\partial * Y$–the set of edges connecting Y to the rest of the graph. The Cheeger Isoperimetric Constant is defined as the infimum of the ratio $\frac{|\partial * Y|}{|Y|}$ over all non-empty finite subsets Y. This measures how difficult it is to cut a finite piece out of the group; a large constant means the "boundary" is always proportionally large.

A group is amenable if and only if its Cheeger Isoperimetric Constant is 0. This means the group contains finite subsets where the boundary is arbitrarily small compared to the set itself. Inversely, a group is non-amenable if and only if its Cheeger Isoperimetric Constant is strictly positive. No matter how you choose a finite subset, a significant fraction of its elements will be on the boundary.

As an example, we can analyze the Cayley graph of F_2, which is an infinite 4-regular tree (each vertex has four edges). In a tree, any finite set of vertices has a boundary that is at least as large as the set itself. A simple ball of radius n has roughly $3 * 4^n$ vertices, but its boundary contains all the vertices at distance $n+1$, which is about $4 * 4^n$ vertices. The ratio $\frac{|\partial * Y|}{|Y|}$ remains bounded away from zero (in fact, it can be shown that it is exactly equal to 1 for standard generators). F_2 is therefore non-amenable.

As for F, through methods such as calculating the density of its Cayley graph (with the relationship density + Cheeger constant = $2m$, where m is the number of generators), the upper bound of the Cheeger Isoperimetric Constant of the Cayley graph of F has been lowered to 0.5.

[3]The proof is out of the scope of this paper. Readers are welcome to check out page 15 of these lecture notes from Solomyak[17].

Doubling Function : A doubling function on a group G (with finite generating set A) is a function $\varphi : G \to G$ such that there exists a constant $K > 0$ where $dist(g, \varphi(g)) \leq K$ for all $g \in G$. The distance is measured in the word metric of the Cayley graph. Every element $h \in G$ has at least two distinct preimages under φ.

Again, we use F_2 as an example. An explicit doubling function is easy to construct. For example, map any reduced word to the two words obtained by prefixing it with a and with b. This moves each element by a distance of 1 and clearly doubles it.

A finitely generated group is non-amenable if and only if it admits a doubling function. If F is non-amenable, a doubling function must exist. The open question would then be: What is the minimal constant K? Research provides a key partial answer: For a 2-generator group, a doubling function with the minimal constant $K = 1$ exists if and only if the Cheeger Isoperimetric Constant of its Cayley graph is greater or equal to 1. With F, we know it has upper bound 0.5. Therefore, if F is non-amenable, any doubling function for it must have a constant $K \geq 2$. The exact minimal K is unknown.

Belk-Brown Sets and Forest Diagrams There is also analysis on the Cayley graph of F using special constructed sets based on forest diagrams of F. Readers are welcome to check out page 31 of Belk's thesis on One-Way Forest Diagrams [1] and page 74 of Burillo's book on Belk-Brown sets [3].

5 Acknowledgements

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